

Karina Maharani

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Portofolio : karinamaharani.github.io

PROFILE

Karina is a penultimate Information Systems student (GPA 3.96/4.00, Top 1% out of 200 students) with an interest in the intersection between researching emerging technologies and industry implementation. Experienced in system design, full-stack development, academic research, and machine learning modelling. She translates research into industry-ready tools to build practical deployment and turn theoretical frameworks into practical products.

PROFESSIONAL EXPERIENCE

Student Researcher, Computer Systems Lab University of Indonesia

2026

- Research Title : Clustered Federated Learning with Low Rank Adaptation for Cross-Silo Healthcare
- Designed and developed a clustered federated learning model integrated with Low-Rank Adaptation (LoRA) to simulate tiered healthcare systems using Flower framework
- Simulated a cross-silo healthcare environment using real-world distributions (WHO data, regional disease patterns)
- Developed dataset pipelines integrating symptoms, diagnosis, and geographic data for healthcare pathway recommendation
- Evaluating model performance to focus on real-world deployment challenges, including non-IID data, system scalability, and model usability in practical healthcare settings

Product Manager Intern, Social Circles Co.

2026

- Improved product flows, UI/UX design, and feature sets, increasing user engagement
- Conducted QA across multiple device architectures to ensure app stability and compatibility
- Ensured project deadlines are met through leading daily stand-up meetings with engineer team
- Integrated OpenAI API to build personalized LLM games, translating AI into product features

Product Innovation and Development Intern, PT Paragon Technology & Innovation

2024

- Researched global and regional beauty and skincare products and markets
 - Conducted market research and competitor analysis for opportunities and consumer preferences
 - Developed decks emerging trends, unique packaging, and innovative formulas for cosmetics
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EDUCATION

Bachelor of Computer Science, University of Indonesia

2023 - Present

- Information Systems Major
- Cumulative GPA: 3.96 / 4.00 (Top 2 of 200 students)
- Expected early graduation (3.5 years)

- Selected as sole Indonesian representative for “The World Nuclear Industry Today” Short Course organized by Tsinghua University in collaboration with the World Nuclear University

RESEARCH INTERESTS

- Computational Modelling and Simulation
- Digital Health Systems
- Federated Learning and Distributed Systems
- Multimodal Learning
- Data and Artificial Intelligence Ethics

TECHNICAL SKILLS

- Programming: Python, Java, TypeScript
- Machine Learning: PyTorch, Scikit-learn, Hugging Face, Federated Learning (Flower)
- Data Processing: Pandas, NumPy, Feature Engineering
- Research Methods: Statistical Analysis, SEM, Experimental Design
- Tools: Git, Docker, Visual Studio Code, Figma

PUBLICATIONS & WORKING PAPERS

- Clustered Federated Learning with Low-Rank Adaptation for Cross-Silo Healthcare (Ongoing, 2026)

PAST PROJECTS

HappyHealthy (available in portfolio site)

- Designed a mobile health concept to support psychiatric medication adherence, centered on drug-drug interaction checking, medication reminders, side-effect logging, and caregiver monitoring with risk alerts
- Specified functional, non-functional requirements, including DrugBank API integration for interaction checking, Database EERD schema, Use-Case Diagram, Use-Case Description, Sequence Diagram
- Conducted early user research, user acceptance utilizing usability testing methodology, and SUS Questionnaire with real end-user

Final Project for Artificial Intelligence and Data Science

- Delivered data-driven business insights of real e-commerce data with high dimensionality and data types
- Built preprocessing and feature engineering pipeline with one-hot encoding, normalization, feature scoring in classification tasks
- Trained and tuned various regression models to select optimal regression models
- Ran clustering experiments across multiple methods and selected Hierarchical Agglomerative using PCA for 2D and 3D visualization

Big Five Personality Theory Correlation towards Adoption of AI Health Assistant (available in portfolio site)

- Conducted a quantitative study evaluating how Big Five personality traits relate to adoption of AI health assistants, including constructs like usefulness, enjoyment, ease of use, perceived risk, and continuance intention
- Performed SEM quality checks such as convergent validity, reliability, and discriminant validity
- Translated results into actionable product implications such as necessity of stronger privacy and security reassurance for high-neuroticism users and exploration-friendly design for high-openness users

OTHER EXPERIENCES

English Facilitator for Asian Medical Students' Association

Depok

- Guided active medical students in focus group discussion regarding Emerging Technology for Healthcare such as genetic editing
- Gave feedback to participants post-debate for technology-centric topic in healthcare and debate conduct

Participant of WHO TDR Implementation Research Program

Jakarta

- Participated in discussions regarding ongoing Implementation Research alongside post-doctoral students
- Developed Implementation Research abstract for E-Health Challenges in Indonesia

Best Delegate of Singapore Model United Nations 2025 UNODA Council

Singapore

- Produced a 50-page research regarding current loopholes, historical accounts, international dynamics, and recent technological breakthroughs regarding the issues and challenges of Autonomous Weapons Systems and Biological Warfare
- Developed a comprehensive framework and solution supported by 40% of committee to prevent misuse of Autonomous Weapons Systems

LANGUAGE CERTIFICATION

- English - IELTS Academic, Overall Band 8.5 (CEFR C2)
- Indonesian - Ujian Kemahiran Bahasa Indonesia, Tingkat Istimewa (CEFR C2); native proficiency